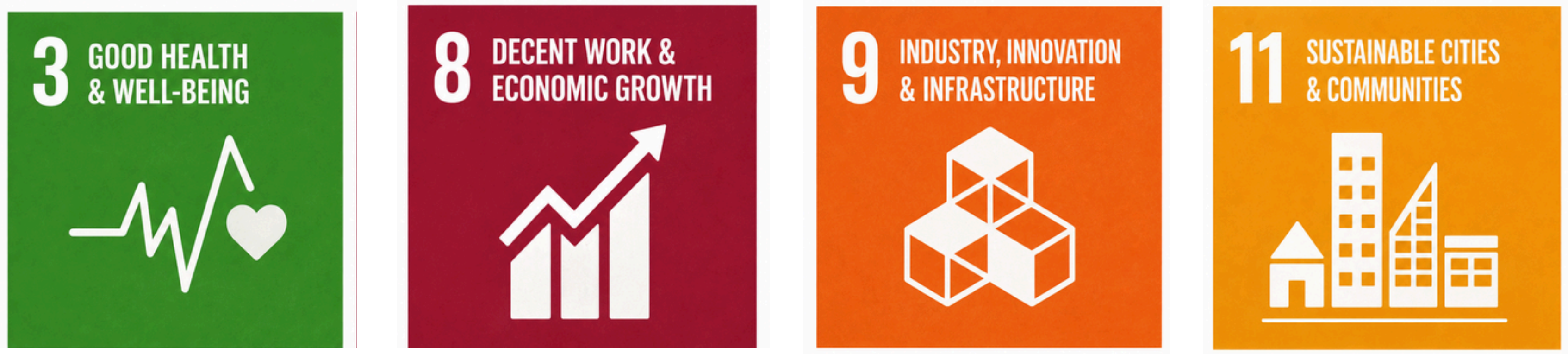
SMART INDIA HACKATHON 2026

KIROSHI

- **Problem Statement ID** – SIH25002
- **Problem Statement Title-** Smart Tourist Safety Monitoring & Incident Response System using Al, Geo-Fencing, and Blockchain-based Digital ID
- **Theme -** Travel and Tourism
- **PS Category-** Software
- **Team Name -** NETRUNNERS
- **Sustainable Development Goals:**



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PROPOSED SOLUTION

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KIROSHI — Keypoint Intelligence for Real-time Observation, Safety & Human Interaction

Digital Tourist ID

Privacy-first digital identity with consent-based access and QR verification.

AI Risk Engine

Detects unusual movement, inactivity, route deviation and high-risk behavior patterns.

Smart Geo-Fencing

Create safe, restricted and high-risk zones with real-time alerts for entry/exit & deviation.

SOS & Incident Response

One-tap SOS, instant incident creation, live location sharing and alerts to authorities.

Trip & Itinerary Management

Plan trips, share itinerary with contacts, check-in at points and track overall progress.

Targeted CCTV Investigation

Search relevant cameras using location & time window with AI detection and re-identification.

Offline SOS Support

Works without internet via local SOS, SMS/relay network and store-and-forward sync.

Authority Dashboard

Real-time map, risk insights, incident management and response coordination.

Blockchain Trust Layer

Tamper-proof records of consent, incident logs and verification hashes stored on blockchain.

Multi-Stakeholder Collaboration

Connects tourists, authorities, emergency teams and local services for faster resolution.

How KIROSHI addresses the problem

Our app addresses tourist safety by combining verified Digital Tourist IDs, trip based GPS monitoring, geo-fencing, AI-assisted risk scoring, and instant SOS. It verifies tourist safety before escalation, provides authorities with location and incident context, and enables targeted CCTV investigation securely.

Innovations

AI Itinerary & Risk Prediction

AI analyzes travel patterns to suggest safe itineraries and predict potential risks.

Intelligent SOS & Incident Verification

Verifies incidents using movement anomaly, geofence and inactivity before escalating an SOS.

Secure Data Sharing & Privacy

Shares real-time data securely with authorities only when necessary, ensuring privacy.

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TECHNICAL APPROACH

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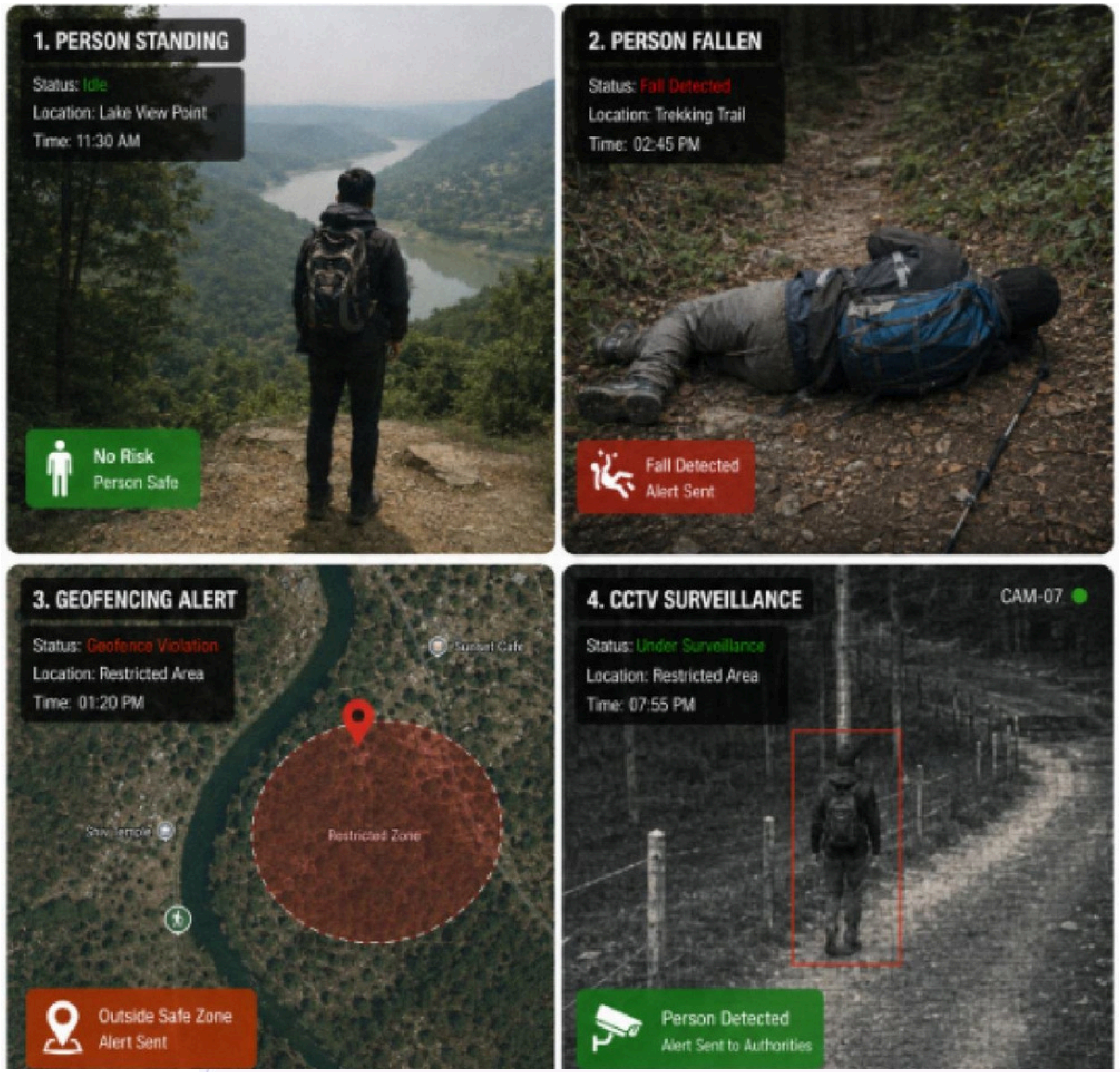

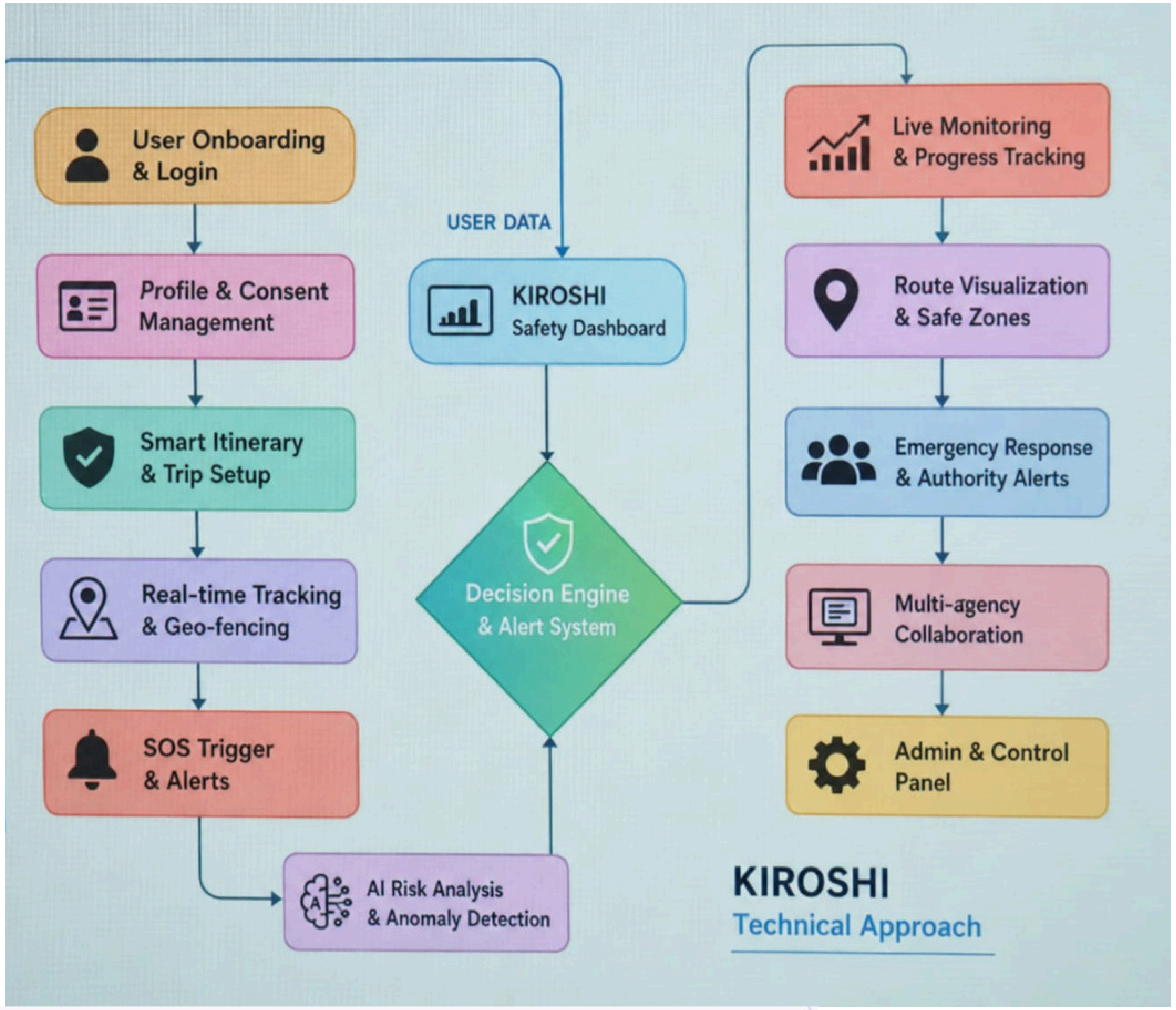

Flutter | Dart | React | Tailwind CSS | FastAPI | Python | PostgreSQL + PostGIS | WebSockets | Mapbox | Scikit-learn | YOLO | OpenCV | Solidity | Ethereum (Testnet)

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FEASIBILITY AND VIABILITY

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FEASIBILITY

TECHNICAL
Uses proven technologies like Flutter, FastAPI, PostGIS, AI/ML, and Blockchain. Open-source and developer-friendly.

MARKET
High demand for tourist safety solutions. Aligned with initiatives like ARAI and Smart Cities.

SOCIAL
Enhances tourist safety, builds trust, and promotes responsible tourism in urban and rural areas.

ECONOMIC
Cost-effective solution. Saves resources through faster incident response.

OPERATIONAL
Easy to integrate with existing authority systems and CCTV infrastructure. Modular design for smooth deployment.

LEGAL & PRIVACY
Privacy-first approach. PII and CCTV remain off-chain and encrypted.

SUSTAINABILITY
Supports long-term safety ecosystem. Encourages safe and sustainable tourism.

1. PERSONAL SAFETY RISK

RISK: Tourists may face accidents, harassment, theft, or get lost in unfamiliar or high-risk areas.

OUR SOLUTION:
- Real-time GPS tracking during active trips
- AI risk scoring, geo-fencing & two-stage verification with SOS

2. DELAYED EMERGENCY RESPONSE

RISK: Delayed alerts and unclear information can slow rescue and increase incident severity.

OUR SOLUTION:
- Instant SOS with location & incident creation
- Real-time alerts, live dashboard & structured incident data

3. PRIVACY & DATA MISUSE RISK

RISK: Personal, location and identity data can be misused or illegally accessed.

OUR SOLUTION:
- Privacy-first design, minimal data collection
- Digital Tourist ID (no PII exposed), blockchain credentials & encryption

4. CONNECTIVITY & NETWORK DEPENDENCY RISK

RISK: Poor or no internet connectivity can prevent alerts or make tourists unable to get help.

OUR SOLUTION:
- Offline SOS via SMS / Bluetooth / LoRa with local storage
- Relay / gateway system to forward alerts & ensure help reaches even in remote areas

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IMPACT AND BENEFITS

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ENHANCED TOURIST SAFETY & CONFIDENCE
Real-time tracking, SOS alerts, fall detection and geo-fencing ensure immediate help and reduce response time.

PERSONALIZED & ENRICHED TRAVEL
AI-powered recommendations, smart itineraries and local insights create seamless and enriching travel experiences.

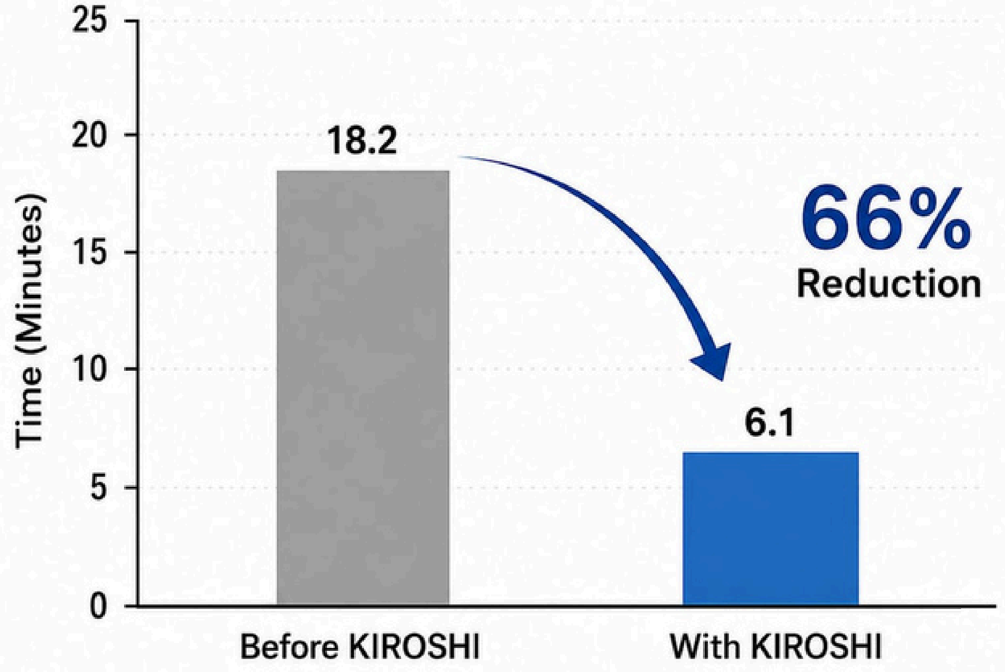
COMMUNITY & ECONOMIC EMPOWERMENT
Promotes local businesses, creates job opportunities and boosts income for local communities.

TRUSTWORTHY & SECURE ECOSYSTEM
Blockchain-based incident logs and secure data handling ensure transparency, accountability and user trust.

NATION BUILDING & GLOBAL IMPACT
Positions India as a leader in smart tourism safety, strengthens its global image and contributes to a secure, progressive and digitally empowered Bharat.

REDUCING RESPONSE TIME, SAVING LIVES

Real-time alerts & smart detection enable faster emergency response



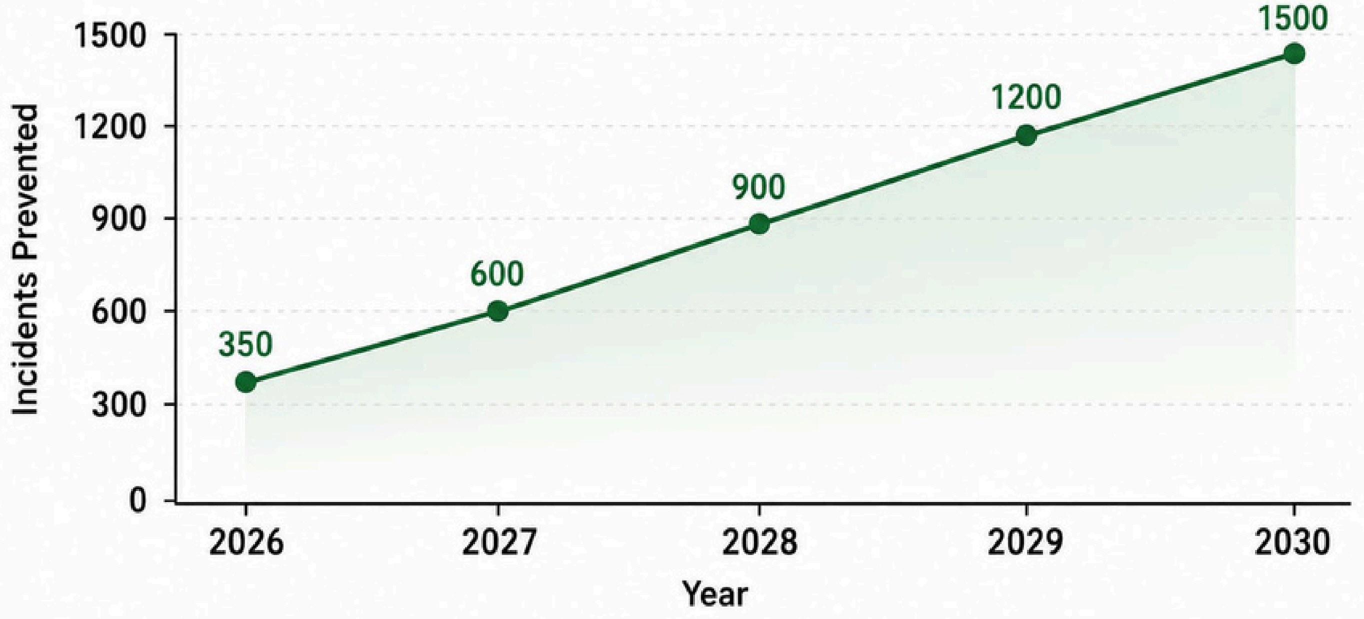
Enhances tourist safety with real-time location monitoring, intelligent risk detection, SOS alerts, and faster coordination between tourists and emergency authorities.
Reduces emergency response time by combining GPS, geo-fencing, AI risk analysis, and a centralized authority dashboard for timely intervention.
Provides proactive tourist protection by identifying unusual movement patterns and enabling authorities to respond before situations become critical.
Builds trust in tourism through privacy-preserving digital identity, secure data handling, transparent incident records, and reliable emergency assistance.

Targeted Beneficiaries

- Tourists (domestic & International)
- Local vendors & small businesses
- Local communities & artisans

ENHANCING TOURIST SAFETY ACROSS INDIA

Potential Incidents Prevented Annually



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RESEARCH AND REFERENCES

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We were inspired by the safety challenges tourists face while exploring unfamiliar, remote, or high-risk destinations. Tourists may experience emergencies, lose connectivity, enter restricted areas, or become separated from their planned routes, making timely assistance difficult. We wanted to create a solution that combines technology with human response rather than relying solely on AI. Our idea uses GPS, geo-fencing, AI-based risk scoring, SOS alerts, digital identity, and an authority dashboard to improve emergency response. Most importantly, we wanted to build a privacy-first system that protects tourist data while helping authorities respond faster and make tourism safer.

MediaPipe Pose is a lightweight, real-time pose estimation technology that detects 33 body landmarks from images or video. It can track body posture and movement from a camera stream, making it suitable for detecting unusual motion patterns and possible falls. KIROSHI can use pose landmarks as one signal alongside GPS, geo-fencing and risk scoring to support faster safety alerts.

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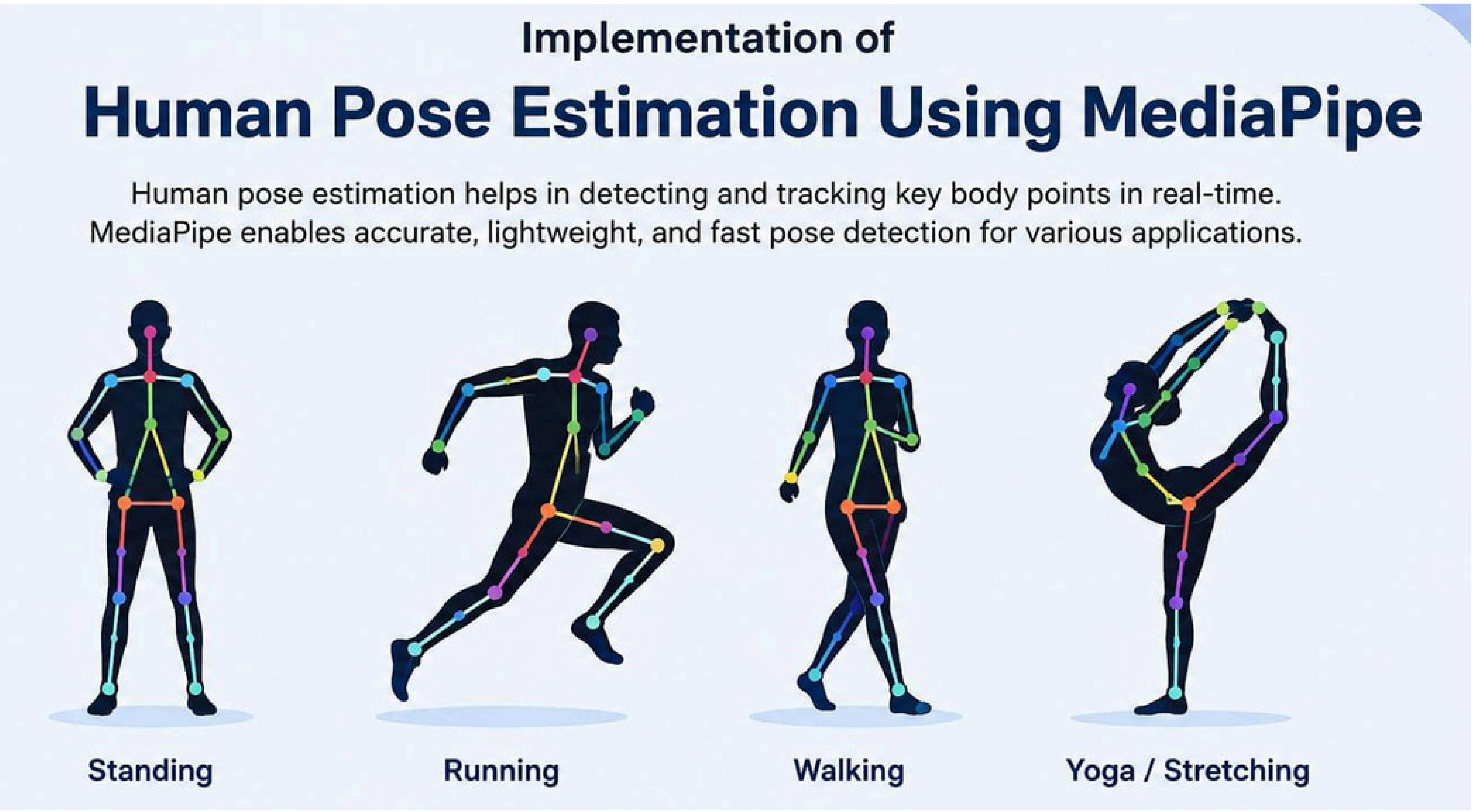

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